

Parte 01

- 1) A saída em formato hexadecimal, separando cada byte:

```
> python3 main.py
Texto limpo: SEGURANCA DA INFORMACAO
Tamanho do texto limpo: 23 bytes
Texto cifrado:
65 e6 99 ae b6 e5 7c e6 33 2e 34 9f cb b3 3a 69 d0 7b ae 4e 3c be e3
Tamanho do texto cifrado: 23 bytes
```

- 2) A saída em formato hexadecimal, separando cada byte. Compare-o com a saída da questão anterior.

Primeira: 65 e6 99 ae b6 e5 7c e6 33 2e 34 9f cb b3 3a 69 d0 7b ae 4e 3c be e3

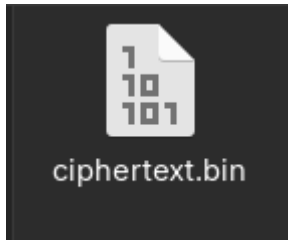
Segunda: 65 e6 99 ae b6 e5 7c e6 33 2e 34 9f cb b3 3a 69 d0 7b ae 4e 3c be e3

Identifique a tag de autenticação.

```
Tag de autenticação:
1b b1 4e 14 fa 1b 19 86 71 e8 48 8b 8c 6c 3e f5
Tamanho da tag de autenticação: 16 bytes
```

Salve o texto cifrado em disco.

```
Texto cifrado salvo no arquivo 'ciphertext.bin'.
```

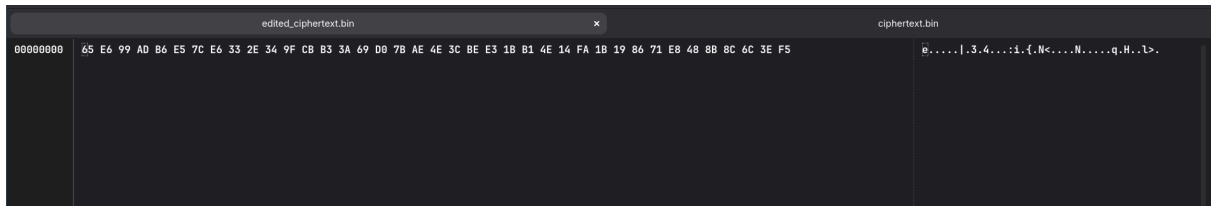


- 3) Exiba o conteúdo do texto decifrado.

```
> python3 main.py
Texto cifrado:
65 e6 99 ae b6 e5 7c e6 33 2e 34 9f cb b3 3a 69 d0 7b ae 4e 3c be e3
Tag de autenticação:
1b b1 4e 14 fa 1b 19 86 71 e8 48 8b 8c 6c 3e f5
Texto decifrado:
SEGURANCA DA INFORMACAO
```

4) Editar um dos bytes e tentar decifrar utilizando os mesmos parâmetros.

Utilizando o programa GHex e alterando um byte (AE -> AD)



Tentativa de decifragem com alteração

```
1 python3 main.py
Texto cifrado:
65 e6 99 ad b6 e5 7c e6 33 2e 34 9f cb b3 3a 69 d8 7b ae 4e 3c be e3
Tag de autenticação:
1b b1 4e 14 fa 1b 19 86 71 e8 48 8b 8c 6c 3e f5
Traceback (most recent call last):
  File "/home/hetzuga/projects/furb/7-semester/info-sec-exes/trabalho-08/parte-01/main.py", line 104, in <module>
    main()
    ~~~~^
  File "/home/hetzuga/projects/furb/7-semester/info-sec-exes/trabalho-08/parte-01/main.py", line 7, in main
    exercise_4()
    ~~~~~^
  File "/home/hetzuga/projects/furb/7-semester/info-sec-exes/trabalho-08/parte-01/main.py", line 98, in exercise_4
    plaintext = cipher.decrypt_and_verify(ciphertext, tag)
  File "/home/hetzuga/projects/furb/7-semester/info-sec-exes/trabalho-08/parte-01/.venv/lib/python3.13/site-packages/Crypto/Cipher/_mode_gcm.py", line 567, in decrypt_and_verify
    self.verify(precalculate_mac_tag)
    ~~~~~^
  File "/home/hetzuga/projects/furb/7-semester/info-sec-exes/trabalho-08/parte-01/.venv/lib/python3.13/site-packages/Crypto/Cipher/_mode_gcm.py", line 508, in verify
    raise ValueError("MAC check failed")
ValueError: MAC check failed
```

Parte 02

1) Criação de usuário salvando senha com hash no banco.

```
tmux 1:nvim 2:yazi
6 ### CREATE ✓ 360.39 ms
1
3 POST http://localhost:8000/users HTTP/1.1
1 Content-Type: application/json
2
3 {
4   "login": "marlon@gmail.com",
5   "password": "senhasegura123"
6 }

Request: 5/5 Code: 0 Duration: 360.39 ms Time: May 23 10:23:39 AM
URL: POST http://localhost:8000/users Env: default Status: 201 Asse
Buffer: requests/create-user.http::1 Name: CREATE

{
  "login": "marlon@gmail.com",
  "password": "$2b$12$YP/mM./jkXx8BTv6VooDZ0Dj92old9e7rIwDvEdEk0zK432C
  "id": 1,
  "created_at": "2026-05-23T13:23:39"
}
```

```
tmux 1:nvim 2:yazi
6 ### CREATE ✓ 371.88 ms
5
4 POST http://localhost:8000/users HTTP/1.1
3 Content-Type: application/json
2
1 {
7   "login": "sofia@gmail.com",
1   "password": "senhasegura123"
2 }

Request: 8/9 Code: 0 Duration: 359.22 ms Time: May 23 10:26:55 AM
URL: POST http://localhost:8000/users Env: default Status: 201 Asse
Buffer: requests/create-user.http::1 Name: CREATE

{
  "login": "sofia@gmail.com",
  "password": "$2b$12$w/0c4Jmzp.oo3ZIffvC.C0pNFyd1Mobiwgy/wihCKw6mQraD
  "id": 2,
  "created_at": "2026-05-23T13:26:55"
}
```

```
tmux 1:nvim 2:yazi
6 ### CREATE ✓ 371.88 ms
5
4 POST http://localhost:8000/users HTTP/1.1
3 Content-Type: application/json
2
1 {
7   "login": "cristina@gmail.com",
1   "password": "senhasegura123"
2 }

Request: 9/9 Code: 0 Duration: 371.88 ms Time: May 23 10:27:01 AM
URL: POST http://localhost:8000/users Env: default Status: 201 Asse
Buffer: requests/create-user.http::1 Name: CREATE

{
  "login": "cristina@gmail.com",
  "password": "$2b$12$Gw.zKp0oI84IKBofef96/.TRR0sE0cusxHcbU822t3sH0cLv
  "id": 3,
  "created_at": "2026-05-23T13:27:01"
}
```

Tabela “users” apresentando senhas salvas com hashes diferentes para cada usuário apesar de terem sido cadastrados com a mesma senha.

```
tmux 1:nvim 2:yazi
Press ? for help
1 SELECT * FROM users;

+ users
+ sec-info-trabalho-7
+ sec-info-trabalho-8 ✓
+ New query
+ Buffers (1)
  » query-2026-05-23-10-19-23 *
+ Saved queries (0)
+ Tables (2)

+ Query results (5)

dbui [-] 8:1 NORMAL sec-info-trabalho-8-query-2026-05-23-10-19-23 LSP: none sql Top 1:19
1 id login password created_at
1 --
2 1 marlon@gmail.com $2b$12$VP/mM./jkXx8BTv6Voo0Z0Dj92oLd9e7rIwDvEdEk0zK432C8Enya 2026-05-23 13:23:39
3 2 sofia@gmail.com $2b$12$w/0c4Jmzp.oo3ZIffvC.C0pNFyd1Mobiwgy/wihCKw6mQraDNzzRa 2026-05-23 13:26:55
4 3 cristina@gmail.com $2b$12$Gw.zKp0oI84iKBofef96/.TRRGsE0cusxHcbU822tS3h0cLvJxHwe 2026-05-23 13:27:01
```

2) Tentativa de login com credenciais inválidas

```
tmux 1:nvim 2:yazi
5 ### LOGIN ✓ 347.19 ms
4
3 POST http://localhost:8000/users/login HTTP/1.1
2 Content-Type: application/json
1
6 {
1 "login": "marlon@gmail.com",
2 "password": "123456"
3 }

<A> Verbose (V) Script Output (O) Report (R) Help (?) ← [ ] →
Request: 6/6 Code: 0 Duration: 347.19 ms Time: May 23 10:24:57 AM
URL: POST http://localhost:8000/users/login Env: default Status: 401
Buffer: requests/login.http::1 Name: LOGIN

{
  "detail": "Invalid email or password"
}
```

Tentativa de login com credenciais corretas

```
tmux 1:nvim 2:yazi
6 ### LOGIN ✓ 347.31 ms
5
4 POST http://localhost:8000/users/login HTTP/1.1
3 Content-Type: application/json
2
1 {
7 "login": "marlon@gmail.com",
1 "password": "senhasegura123"
2 }

<A> Verbose (V) Script Output (O) Report (R) Help (?) ← [ ] →
Request: 7/7 Code: 0 Duration: 347.31 ms Time: May 23 10:25:39 AM
URL: POST http://localhost:8000/users/login Env: default Status: 200
Buffer: requests/login.http::1 Name: LOGIN

{
  "status": "success!"
}
```